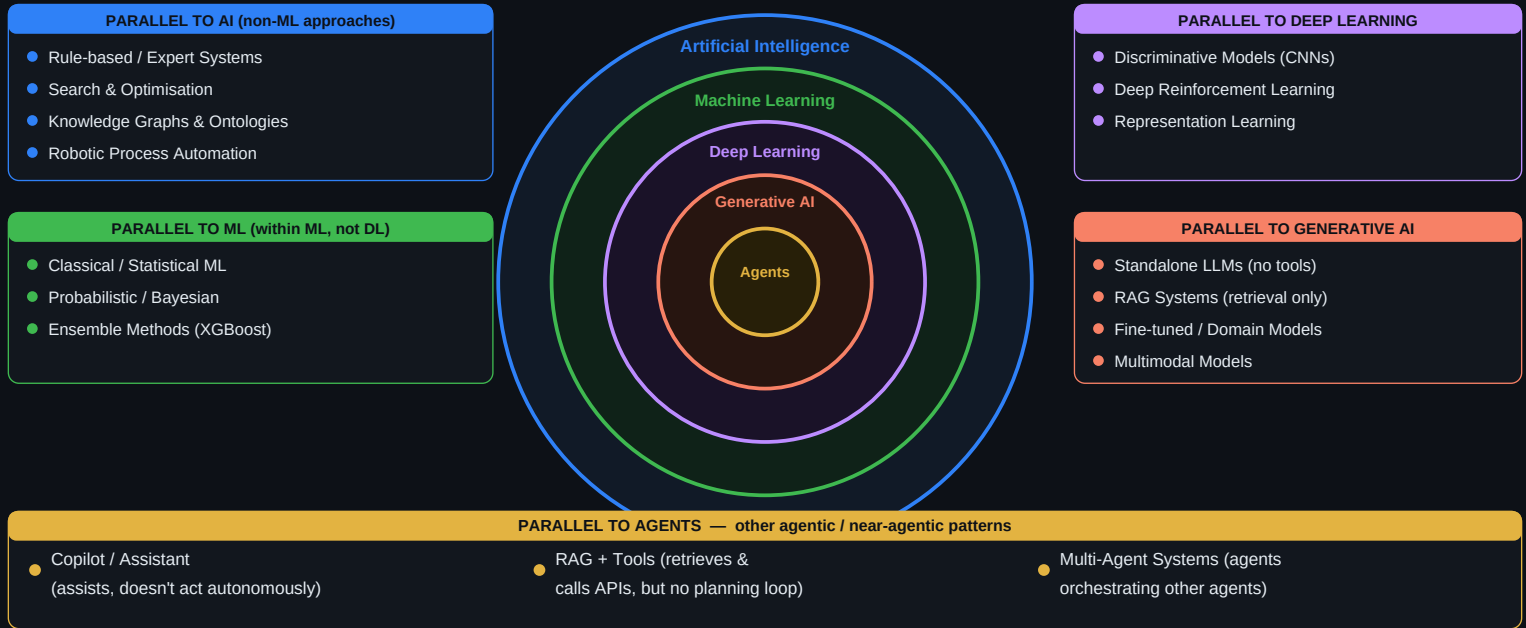


The Complete AI Landscape

A Strategic Reference Map · AI Transformation Journey · Week 1 · Page 1 of 2

① THE AI FAMILY TREE

Each inner ring is a specialisation of the one outside it — plus what sits parallel at each level



The Complete AI Landscape

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② THE FOUR ML PARADIGMS

How machines learn — the fundamental approaches underneath every AI product

01 SUPERVISED

- Learns from labelled input/output pairs
- Classification & regression tasks
- Most common in enterprise AI
- Requires clean, labelled datasets

e.g. fraud detection, credit scoring

02 UNSUPERVISED

- No labels — finds hidden structure
- Clustering, segmentation, anomalies
- Dimensionality reduction
- Good for exploration & discovery

e.g. customer segments, outlier detection

03 REINFORCEMENT

- Agent learns via reward & penalty
- Trial and error over episodes
- Optimises long-run strategy
- Powers game AI, robotics, trading

e.g. AlphaGo, trading bots

04 SELF-SUPERVISED

- Creates its own labels from raw data
- Foundation of modern LLMs
- Predict next token / masked word
- Scales to internet-size datasets

e.g. GPT, Claude, BERT

③ AI CAPABILITY TYPES — What the system does

The strategic lens: match the business need to the right AI capability

05 PERCEPTION

- Computer vision
- Speech recognition
- Document OCR
- Sensor processing

ID verify, defect detect

06 PREDICTION

- Forecasting values
- Risk & credit scoring
- Demand & churn
- Time-series

Loan default, sales fcast

07 GENERATION

- Text, code, images
- Summarisation
- Translation
- Synthetic data

Report drafting, code

08 REASONING

- Multi-step logic
- Document Q&A
- Decision support
- Agent planning

Regulatory, RAG

09 RECOMMEND.

- Ranking & personalise
- Collaborative filter
- Next-best-action
- Search ranking

Product recs, policy

10 OPTIMISATION

- Scheduling & routing
- Portfolio allocation
- Scenario planning
- Ops research

Logistics, capital alloc

④ HOW AI IS DEPLOYED — The product form it takes

Same model, very different product — the form determines adoption, risk, and value

EMBEDDED AI

Silent auto-decisions in background

Fraud alerts, autocomplete

ANALYTICS & BI

Insight surfaces in dashboards

Risk dashboards, KPI fcast

COPILOT / ASSIST

Helps human complete tasks faster

Code assist, doc drafting

AUTONOMOUS AGENT

Acts independently using tools

Research agent, monitor

PLATFORM / API

AI as shared infrastructure layer

OpenAI, Anthropic, Azure

⑤ STRATEGIC DECISION GUIDE — From client problem to AI approach

Your consulting translation layer: what the client says → what AI branch to explore

What the client says	AI archetype	Technical approach
"We want to automate document review"	Document Intelligence	Supervised ML + LLM extraction
"We want to reduce fraud losses"	Prediction + Anomaly	Supervised / Unsupervised ML
"We want a chatbot for staff queries"	Knowledge Assistant	RAG + LLM (Generative AI)
"We want to personalise recommendations"	Recommendation Engine	Collaborative filtering / ML
"We want AI to monitor our operations"	Monitoring Agent	Agents + Anomaly Detection
"We want to optimise our supply chain"	Optimisation	Operations Research + ML